PDF Signing API User Guide

Staging environment

Base URL https://pdf-signer-staging.dswd.gov.ph

This guide is for you if you want to use the DSWD PDF Signing API from Postman or connect it to your application. It walks you through setting up your project folder, uploading a certificate, signing a PDF, and checking or removing your saved certificate.

The examples use sample names, passwords, and file locations. Replace them with your own values. The request formats in this guide reflect the repository code reviewed on October 6, 2026. Choose your own unique project folder before the first upload and keep it for all requests from the same application and environment.

1 What this service does

An API is a way for one application to ask another application to do something. Here, your application sends a PDF and signing instructions. The service processes the request and sends back the resulting PDF.

You can choose between two signature options:

Option	What it does	What you need
Digital signature (`digital`)	Signs the PDF using a certificate and private key. A compatible PDF reader can inspect the signature and report changes to the document.	A valid signing certificate and its password.
Saved image signature (`electronic`)	Places the signature image saved in the user's PNPKI profile onto the PDF. It does not add a certificate-based digital signature.	A saved signature image and a position for it. No certificate password is needed for this request.

The service also lets you store certificates for later use, view their details, check whether they have been revoked, and remove an active certificate.

2 A few terms you will see

Term	Simple meaning
Endpoint	The address for a particular action, such as `/sign`.
Request	The information you send to the service.
Response	The result the service sends back.
Header	An extra piece of information sent with a request, such as a project key.
Body	The main part of a request, containing fields and uploaded files.
Vault	The server's encrypted storage for certificates and signature images.
Certificate	A file containing the signer's certificate identity. Signing also needs its matching private key.

Term	Simple meaning
PKCS#12 / `.p12`	A commonly used password-protected package containing a certificate and private key.
PEM	A text-based format for certificate and key material. A signing bundle must include the private key and certificate chain.
Revoked	The certificate has been withdrawn by its issuing authority.
Project key	A name that separates one application's saved users from another application's users.
User ID	Your application's identifier for the person whose certificate or image is saved.

3 Before you begin

Use the staging base URL https://pdf-signer-staging.dswd.gov.ph. Before your first request, complete these two steps:

1. Send a health check to confirm that you can reach the staging API.
2. A unique project folder for your application and environment, for example idms-staging-dadkalakdqwe.

Prepare a PDF, the user's certificate file and password, and optionally a signature image. A clear PNG with a transparent background is a practical choice for the image.

The staging base URL is `https://pdf-signer-staging.dswd.gov.ph`. Add the endpoint to this address. For example, the signing address is `https://pdf-signer-staging.dswd.gov.ph/sign`. The word staging identifies the environment used for testing integrations.

Keep certificate passwords in your application's protected server settings. Use HTTPS when sending credentials over a network. Your application needs to control which person may request a signature.

Choose a unique project folder before your first request

Your projectFolder must be unique to your application and environment. Do not reuse an existing project folder such as IDMS. This keeps your uploads separate from other projects and avoids replacing another project's active certificate when the user IDs are the same.

Use an application name and environment followed by a random identifier, UUID, or hash. Examples are idms-staging-dadkalakdqwe, idms-staging-7f9c2a81e4b6430da65c817f902a6d3e, or idms-staging-550e8400-e29b-41d4-a716-446655440000. Use simple letters, numbers, and hyphens.

These examples are placeholders. Generate a new random value for your integration rather than copying a folder used by another application or team. Save it in your application settings and Postman environment. A random name reduces the chance of a collision; it is not a password or an access control.

Generate the folder once during setup, not for every request. Use the exact same value for projectFolder and X-Project-Key when uploading, reading details, checking status, signing, and removing a certificate. Keep separate values for staging and production.

All request examples below use idms-staging-dadkalakdqwe. Replace it everywhere with your own unique folder before using them.

Project headers

For all examples using projectFolder=idms-staging-dadkalakdqwe, send:

```
X-Project-Key: idms-staging-dadkalakdqwe
```

The project header must match the project value in the request. For example, idms-staging-dadkalakdqwe in the header must match idms-staging-dadkalakdqwe in projectFolder.

The API also accepts `projectKey`, `project`, and `projectName` as alternatives to `projectFolder`. Use just one consistently. Project scope is optional, but a request without a project value must also omit `X-Project-Key`.

Keep the user and project consistent

If you upload with projectFolder=idms-staging-dadkalakdqwe and userId=user-1, use those same values when reading, signing, checking, or removing that record. The response may display idms-staging-dadkalakdqwe/user-1; continue sending user-1 as the user ID with idms-staging-dadkalakdqwe as the separate project value.

4 Available actions

Method and endpoint	Short description	Successful result
`GET /health`	Check whether the service responds.	JSON with `status: ok`.
`POST /certificates/upload`	Save a certificate and optional signature image.	Certificate details in JSON.
`GET /certificates/info`	View the active saved certificate's details.	Certificate details in JSON.
`POST /certificates/status`	Check the saved certificate's revocation status.	Certificate details and status in JSON.
`POST /sign`	Add a digital signature or saved image to a PDF.	A PDF file.
`DELETE /certificates/remove`	Remove an active saved certificate.	JSON confirming removal.

JSON is a text response made of named values, such as `{"status":"ok"}`. Signing returns the PDF itself, rather than JSON.

5 Set up a request in Postman

For each action, create a request and choose the method from the table above. Enter the full address, such as `https://pdf-signer-staging.dswd.gov.ph/sign`.

Add the project headers in the Headers tab. Then choose the body type described for that action:

Body type	When to use it	How to enter values
`form-data`	Uploading a certificate or PDF.	Use Text for ordinary fields and File for attachments.
`x-www-form-urlencoded`	Checking certificate status, removing a certificate, or signing with server file paths.	Enter each field as a text value.
No body	Health and certificate information requests.	Put any required query values in Params.

For form-data, let your client generate the multipart Content-Type header, including its boundary. Do not type a bare `multipart/form-data` header yourself.

The command examples use curl.exe and PowerShell's backtick for continuation. Keep the backtick at the very end of its line. Use Postman for entering real credentials if you want to avoid putting them in shell history.

6 Check that the service is available

GET `/health`

Send this request first. It needs no body.

```
curl.exe "https://pdf-signer-staging.dswd.gov.ph/health"
```

Expected response:

```
{"status":"ok"}
```

This confirms that the service responds. It does not confirm that your certificate, password, or signature placement will work.

7 Save a certificate and signature image

POST `/certificates/upload`

Uploading once lets you sign later by sending the user ID instead of attaching the certificate again.

In Postman, choose Body > form-data:

Field	Type	Required	Short description
`projectFolder`	Text	If using project scope	Your unique project folder. Choose it once and reuse it.
`userId`	Text	Yes	The owner of this saved record.
`password`	Text	Yes	The certificate password; also used when protecting stored material.
`p12File`	File	One certificate source is required	The user's PNPKI `.p12` file.
`signatureImageFile`	File	No	The signature image to save for later requests.

Example:

```
curl.exe -X POST "https://pdf-signer-staging.dswd.gov.ph/certificates/upload" `
  -H "X-Project-Key: idms-staging-dadkalakdqwe" `
  -F "projectFolder=idms-staging-dadkalakdqwe" `
  -F "userId=user-1" `
  -F "password=YOUR_CERTIFICATE_PASSWORD" `
  -F "p12File=@C:/certificates/your-pnpki.p12" `
  -F "signatureImageFile=@C:/documents/signature.png"
```

The `@` tells curl to upload the file from your computer.

Other certificate sources are `pemFile`, `crtFile`, `p12Path`, `pemPath`, and `crtPath`. Send only one certificate source. The CRT option is read as PEM; a public certificate alone is not enough for signing because the matching private key is needed.

You can use `signatureImagePath` instead of `signatureImageFile`. Send only one image source during upload.

Fields ending in `Path` refer to files already on the signing server. A file path on your own laptop will not work against a different server unless that file exists there too.

Example response with illustrative values:

```
{
  "userId": "idms-staging-dadkalakdqwe/user-1",
  "certificateSubject": "CN=Juan Dela Cruz,O=Department of Social Welfare and Development,C=PH",
  "certificateIssuer": "CN=Government Signing CA - G2,O=DICT,C=PH",
  "certificateSerial": "123456789",
  "certificateExpiry": "2027-02-19T03:29:37Z",
  "isActive": true,
  "createdAt": "2026-10-06T01:00:00Z"
}
```

Response field	Short description
`userId`	The saved user's identifier, including the project when provided.
`certificateSubject`	The identity written in the certificate.
`certificateIssuer`	The certificate's issuing authority.
`certificateSerial`	The certificate's identifying serial number.
`certificateExpiry`	When the certificate expires.
`isActive`	Whether this is the active vault record. This does not mean it has passed a fresh revocation check.
`createdAt`	When the vault record was created.

Times ending in `Z` are in UTC. Add eight hours to read them in Philippine time.

Uploading another certificate for the same project and user makes the new upload active and archives the previous record. Include the image in the new upload if you want it saved with the replacement record.

8 View saved certificate details

GET `/certificates/info`

Use this to confirm which certificate is currently saved. No certificate password is required for this endpoint.

In Postman, put `projectFolder` and `userId` in Params, and add the matching project headers.

```
curl.exe --get "https://pdf-signer-staging.dswd.gov.ph/certificates/info" `
  -H "X-Project-Key: idms-staging-dadkalakdqwe" `
  --data-urlencode "projectFolder=idms-staging-dadkalakdqwe" `
  --data-urlencode "userId=user-1"
```

The result contains the same certificate detail fields shown in the upload response. This endpoint reads stored information; use the status endpoint for a revocation check.

9 Check certificate status

POST `/certificates/status`

Use this when you want to check whether the issuing authority has withdrawn a saved certificate.

Choose Body > x-www-form-urlencoded. Send `projectFolder`, `userId`, and `password` as text fields.

```
curl.exe -X POST "https://pdf-signer-staging.dswd.gov.ph/certificates/status" `
  -H "X-Project-Key: idms-staging-dadkalakdqwe" `
  --data-urlencode "projectFolder=idms-staging-dadkalakdqwe" `
  --data-urlencode "userId=user-1" `
  --data-urlencode "password=YOUR_CERTIFICATE_PASSWORD"
```

Alongside the certificate details, the response includes:

Field	Short description

Field	Short description
`revocationStatus`	`GOOD`, `REVOKED`, or `UNKNOWN`.
`revocationMessage`	An explanation of the result.
`revocationCheckedAt`	When the check was made.
`revocationSource`	The source used by the check.

`GOOD` means the revocation check passed. `REVOKED` means the certificate was confirmed withdrawn. `UNKNOWN` means the service could not establish a definite answer; it does not mean the certificate is valid or revoked.

The service checks revocation using the certificate information and trusted issuer information available to it. If it cannot establish a definite result, you may receive UNKNOWN. Read the revocation message before deciding how your application should handle that result.

10 Digitally sign a PDF using a saved certificate

POST `/sign`

Choose Body > form-data. The usual fields are:

Field	Type	Required	Short description
`projectFolder`	Text	If using project scope	The project used during upload.
`userId`	Text	Yes for vault signing	The user used during upload.
`password`	Text	Yes for digital signing	The saved certificate's password.
`signatureMode`	Text	No	Use `digital`; this is also the default.
`pdfFile`	File	Yes when uploading a PDF	The document to sign.
`pageIndex`	Text	No	Page number starting from zero.
`x`, `y`, `width`, `height`	Text	No for digital signing	Where the visible signature goes and how large it is.
`fieldName`	Text	No	A name for this digital signature field.

Example:

```
curl.exe --fail-with-body -X POST "https://pdf-signer-staging.dswd.gov.ph/sign" `
  -H "X-Project-Key: idms-staging-dadkalakdqwe" `
  -F "projectFolder=idms-staging-dadkalakdqwe" `
  -F "userId=user-1" `
  -F "password=YOUR_CERTIFICATE_PASSWORD" `
  -F "signatureMode=digital" `
  -F "pdfFile=@C:/documents/input.pdf" `
  -F "pageIndex=0" `
  -F "x=120" -F "y=80" -F "width=210" -F "height=90" `
  -F "fieldName=Signature1" `
  -F "signerName=Juan Dela Cruz" `
  -F "signerLocation=Philippines" `
  -F "signReason=Approved" `
  -o "C:/documents/signed.pdf"
```

On success, the body is a PDF. In Postman, save or download the response to a `.pdf` file. In curl, `-o` saves the response on your computer. Check the HTTP result before treating the downloaded file as a PDF: an error response is JSON and may also be written to that output filename.

Open the result in a PDF reader that supports signature checking. The visible name and image are appearance settings; the certificate provides the identity used for cryptographic signing.

Signing without saving the certificate first

For a one-time digital signing request, replace `userId` with `p12File` and attach the certificate. Keep `password` and the PDF fields. Do not send both a vault user and a direct certificate.

Example replacement field:

```
p12File: choose your-pnpki.p12 as a File
```

Other direct options are `p12Path`, `pemFile`, `crtFile`, `pemPath`, and `crtPath`. Use one source. A PEM bundle needs a private key and certificate chain; the digital request parser still requires a nonblank `password` field even though PEM loading does not use it to unlock an encrypted PEM key.

Choosing a signature image for digital signing

The service uses an uploaded `signatureImageFile` first, then `signatureImagePath`, then the saved vault image, and finally its fallback image. Usually, leaving out the image fields lets it use the image you already saved. Send just one image override when needed.

11 Add the saved signature image only

POST `/sign` with `signatureMode=electronic`

Use this option when you need the user's saved signature image placed on a PDF. The request does not access the certificate's private key or add a cryptographic signature.

First save the image through the certificate upload endpoint. There is no separate image-only upload endpoint in this API.

```
curl.exe --fail-with-body -X POST "https://pdf-signer-staging.dswd.gov.ph/sign" `
  -H "X-Project-Key: idms-staging-dadkalakdqwe" `
  -F "projectFolder=idms-staging-dadkalakdqwe" `
  -F "userId=user-1" `
  -F "signatureMode=electronic" `
  -F "pdfFile=@C:/documents/input.pdf" `
  -F "pageIndex=0" `
  -F "x=120" -F "y=80" -F "width=210" -F "height=90" `
  -o "C:/documents/image-signed.pdf"
```

For this option:

- `userId` and an existing saved image are required.
- `password` is not required.
- Supply `x`, `y`, `width`, and `height`. Positions must be nonnegative and dimensions must be positive.
- Request-level image files or paths do not replace the saved image in this mode.
- The image is fitted inside the box while keeping its proportions.
- Digital appearance settings such as signer labels, date text, field reservations, and rotation do not apply to the image-only stamp.
- The service rejects image-only signing if the PDF already contains a digital signature. Apply image-only signatures before digital signatures.

12 Put the signature in the right place

The position uses PDF points, rather than screen pixels. On a typical unrotated page, `x` starts from the left and `y` starts from the bottom. There are 72 points in one inch.

Setting	Example	Short description

Setting	Example	Short description
pageIndex	0	First page. Use 1 for the second page, 2 for the third, and so on.
x	120	Distance from the left of the page.
y	80	Distance from the bottom of the page.
width	210	Width of the signature box.
height	90	Height of the signature box.

Start with one test document, save the output, and adjust the numbers until the signature sits in the intended space. Rotated or cropped PDFs can make screen coordinates differ from PDF coordinates, so check the actual output.

Extra appearance options for digital signing

Field	Short description
signerName	Name shown in the visible appearance. It does not change the certificate owner.
signerLocation	Location recorded for the signature.
signReason	Reason recorded for the signature, such as Approved.
showDate	Show the date with true, or hide it with false. Default is true.
signatureLayout	Choose LEFT_IMAGE or TOP_IMAGE for the image and text arrangement.
footerAlignment	Footer position: LEFT, CENTER, or RIGHT. Default is RIGHT.
labelFontSize	Size of the label text in points.
nameFontSize	Size of the name text in points.
dateFontSize	Size of the date text in points.
rotationDegrees	Unified rotation setting. Use 0, 90, 180, or 270. Overrides the separate rotation settings.
signatureRotationDegrees	Rotate the signature content.
imageRotationDegrees	Rotate only the image content.
rectangleRotationDegrees	Rotate the signature rectangle by 0, 90, 180, or 270.

Leave advanced settings out until the basic placement works.

Repeat the appearance on other parts of the PDF

Use additionalSignaturePlacements for more visible placements in the same request. Put one placement on each line in this order:

```
pageIndex,x,y,width,height
```

For example, enter the following as one multiline Text field in Postman:

```
0,120,180,210,90
1,120,80,210,90
```

These add an appearance on the first and second pages. In digital mode, repeated appearances represent the same cryptographic signing operation; they do not mean additional people signed. In electronic mode, they repeat the saved image.

13 Let more than one person digitally sign

Each person signs the PDF returned by the previous signing request:

3. The first person signs `original.pdf`, producing `step1.pdf`.
4. The second person signs `step1.pdf`, producing `step2.pdf`.
5. The third person signs `step2.pdf`, producing `step3.pdf`.

Use the correct user's ID and password for each step. If you send the original PDF again, the new output will not contain the signatures added to the previous output.

Before the first digital signature, reserve the visible fields for the planned signers with `reservedSignatureFields`. Each line uses:

```
fieldName,pageIndex,x,y,width,height
```

Example first-request value:

```
Signature1,0,120,80,210,90
Signature2,0,120,180,210,90
Signature3,0,120,280,210,90
```

Use `fieldName=Signature1` for the first signer, `Signature2` for the second, and `Signature3` for the third. Later requests reuse the reserved field rather than creating another visible field. Reserving the fields early helps avoid changing the page's annotations after the first signature.

Keep input and output files separate. Check the signature count after each request, and inspect all signatures in a compatible PDF reader. See the incremental signing guide for the existing Postman workflow.

14 Sign files already on the server

If the PDF is already on the signing server, send `inputPdf` instead of `pdfFile`. Use Body > x-www-form-urlencoded when the request contains only text fields.

Example fields for vault signing:

```
projectFolder=idms-staging-dadkalakdqwe
userId=user-1
password=YOUR_CERTIFICATE_PASSWORD
signatureMode=digital
inputPdf=C:/documents/input.pdf
outputPdf=C:/documents/server-signed.pdf
```

Add the same project headers. `outputPdf` is optional: it asks the service to keep the result at that server location. The PDF is still returned in the response.

`outputPdf` saves on the server. Curl's `-o` saves the response on the caller's computer. If `outputPdf` is omitted, the service uses a temporary output that is cleaned up after handling the request. Your application should save the returned PDF if it needs a lasting copy.

15 Understand the signing response

A successful signing request returns HTTP `200` with Content-Type `application/pdf`.

Response header	Short description
`X-Signature-Mode`	`digital` or `electronic`.

Response header	Short description
`X-Signature-Count`	Total digital signatures found in the output PDF. It counts digital signatures, not visible images.
`X-Signed-File-Size`	Output PDF size in bytes.
`X-Request-ID`	An identifier you can include when reporting a signing problem.

An image-only result normally has a digital signature count of `0`. Your application may send its own `X-Request-ID` to help connect its logs with the signing service's logs.

16 Remove an active certificate

```
curl.exe -X DELETE "https://pdf-signer-staging.dswd.gov.ph/certificates/remove?projectFolder=idms-staging-dadkalakdqwe&userId=user-1" `
  -H "X-Project-Key: idms-staging-dadkalakdqwe" `
  --data-urlencode "password=YOUR_CERTIFICATE_PASSWORD"
```

DELETE `/certificates/remove`

Use this when the active saved certificate should be removed. The user ID and project go in the URL query, while the password goes in the form-encoded body.

In Postman:

6. Choose DELETE and enter `/certificates/remove`.
7. In Params, add projectFolder=idms-staging-dadkalakdqwe and userId=user-1.
8. Add the project headers.
9. Choose Body > x-www-form-urlencoded and add `password`.

Expected success response:

```
{"success":true,"removed":true}
```

The password is verified before removal. An expired certificate can still be removed. If no active record exists, the endpoint returns HTTP `404`. Removal archives the active record into history, so it becomes unavailable as the active signing profile. It does not permanently erase historical copies or backups.

17 Common problems and what to do

Errors are returned as JSON, usually in this form:

```
{"error":"Missing required parameter: password"}
```

HTTP code or problem	What it means	What to do
`400`	A field, credential, file, or signing instruction was not accepted.	Read the error message and check the request fields.
`404` during removal	There is no active certificate to remove for that user.	Check the project and user ID.
`405`	The endpoint was called with the wrong method.	Use GET, POST, or DELETE as shown in this guide.
`413`	The complete request is too large.	Reduce attachment sizes. If the request still fails, contact API support with the total request size.

HTTP code or problem	What it means	What to do
`500`	The server could not complete the request.	Report the endpoint, time, error message, and signing request ID to API support. Do not include passwords or private keys.
Project header mismatch	The header and request project differ, or one is missing.	Make both project values match.
Wrong certificate password	The certificate or vault material could not be unlocked.	Use the password associated with that certificate upload.
Multiple signing sources	The request includes a vault user and direct certificate material together.	Choose one signing source.
Certificate expired or not yet valid	The certificate is outside its validity period.	Use a currently valid certificate.
Issuer rejected	The certificate does not meet the service's issuer rules.	Use the required DICT-issued PNPKI signing certificate.
Certificate revoked	The revocation check confirmed withdrawal.	Arrange a valid replacement certificate.
Saved image missing	Electronic mode has no saved image for this user.	Upload the signature image with the user's certificate profile.
Image-only request on a digitally signed PDF	Adding page content would risk the existing digital signature.	Apply image-only signatures before digital signatures.
Server file not found	The supplied path does not exist or cannot be read on the server.	Upload the file, or correct the server-side path and permissions.

For form-encoded requests, use Postman's field editor or curl's `--data-urlencode`, especially when passwords contain &, +, or other special characters.

The default complete-request limits are 25 MiB for /sign, 10 MiB for certificate upload, and 64 KiB for certificate status and removal bodies. The staging service may use different signing and upload limits. Limits include the request's fields and attachments, not just the PDF.

18 Certificate rules in this service

Certificate upload and digital signing reject expired certificates, certificates that are not yet valid, and certificates whose leaf issuer does not match all of these attributes:

```
C=PH
O=DICT
CN=Government Signing CA - G2
```

These issuer checks are mandatory.

Digital signing also checks revocation and blocks a confirmed `REVOKED` result. The current implementation allows signing when revocation is `UNKNOWN`, so a successful request does not prove that a definitive revocation check was available.

Use your own appropriate, valid PNPKI certificate when testing digital signing. An unrelated sample certificate may not meet the issuer policy.

19 Your usual signing workflow

10. Check `/health` to confirm the service is reachable.
11. Upload the user's certificate and signature image when setting up or replacing their profile.
12. Read `/certificates/info` to confirm the active certificate.
13. Choose digital signing or the saved image option for the document.
14. Send `/sign` and save the returned PDF.
15. Open the PDF to check the placement and, for digital signing, inspect its signatures.
16. If another person needs to digitally sign, send the latest signed PDF in the next request.